Diagnosing Agent Failures at Scale

Using an LLM judge to automatically label why AI agents fail

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RESULT *0.5573* → **0.7405** **Hierarchical judging closes most of the gap between a single-pass LLM judge and human annotation.**

micro-F1 on BFCL hold-out · 35 GPT-5 traces · 142 label instances · macro-F1 also improves from 0.52 to 0.62

PROBLEM STATEMENT

Benchmark scores alone don't help us understand *why* agents are failing.

1 An **agent** is an LLM (e.g., GPT-5) wired up to tools such as Gmail, GitHub, or a file system, so it can take actions on the user's behalf instead of just chatting.

2 When the agent fails, the benchmark only reports `valid: False`. That tells us nothing about the cause. Was the wrong tool called? Was a step skipped? Did the agent invent a parameter? Each case needs a different fix.

REAL FAILURE · BFCL · GPT-5

```
user: "Write the statistics to Annual_Report_2023.docx: Earning: 2000, Expenditure: 500, Name: Gorilla"
agent: reorders fields, writes them on separate lines
↳ valid: False · instance_state_mismatch
```

The agent finished. Just not exactly as required. "Failed" is the wrong word; "completed but did not match" is the right one.

3 The AFT source annotations hand-read 500+ such failures and built the **Agent Failure Taxonomy (AFT)**, a hierarchical vocabulary of 19 failure categories. Hand-annotation does not scale. **We built an LLM judge** that reproduces those labels automatically across the 15 categories we focus on (we exclude benchmark-bug labels for now).

TAXONOMY DEFINITIONS

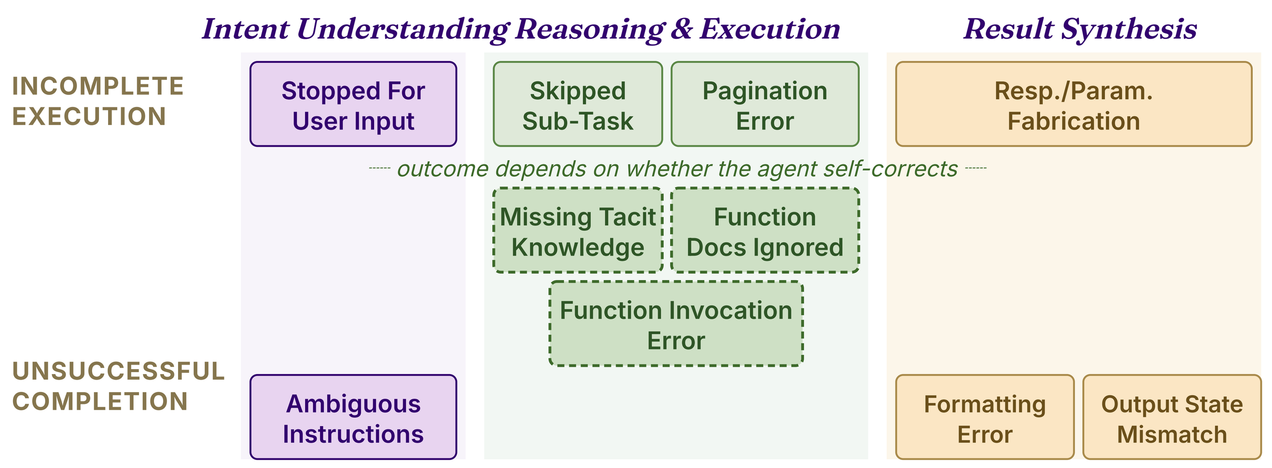
AFT categorizes all agent failures into 19 distinct failure categories.

AFT organizes failures along two dimensions: **completion status** (did the agent finish?) and **workflow phase** (intent, reasoning, or output). Our *Annual Report* case sits in the bottom-right cell, an **Output State Mismatch**.

EXAMPLE FAILURE TYPE

Missing Tacit Knowledge

The agent lacked basic knowledge the task assumed but did not state — e.g., asked to "cancel my booking," it never looked up the user's booking ID first.

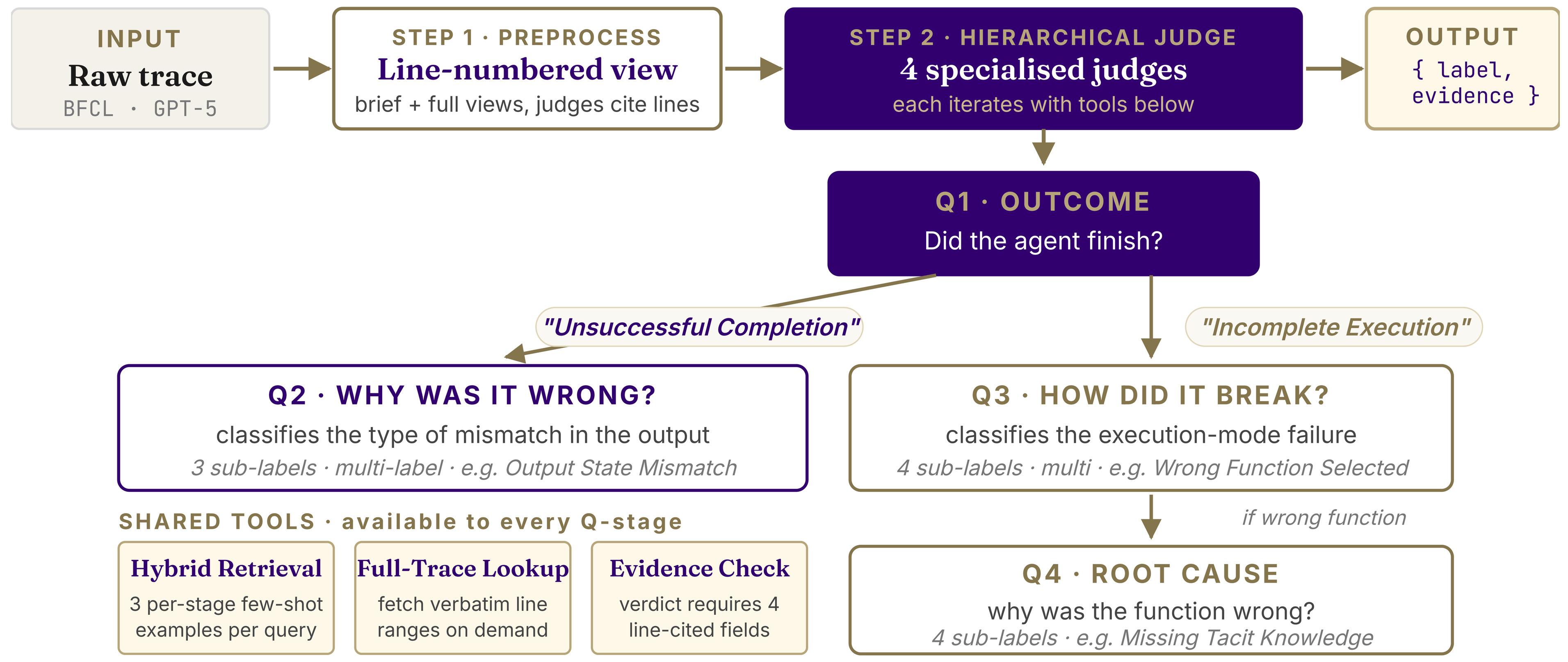


AFT is the answer key. *Our judge is graded on how well its labels agree with the 500+ human annotations.* Figure adapted from AFT taxonomy materials.

PIPELINE

Hierarchical Classification Pipeline

One model assigning 15 overlapping labels at once is hard. We split the decision into a chain of narrow questions, each with its own prompt, examples, and a shared toolset for fetching evidence.



RESULTS · TWO EXPERIMENTAL AXES

Architecture + Q3 prompt design.

Two independent levers: one changes the judge's *structure*, the other the *prompt* for a single stage. Each row adds one component to the row above.

AXIS 1 — ARCHITECTURE STACK (MICRO-F1)

CONFIGURATION	F1	Δ
Single-pass judge	0.557	–
+ Random few-shot	0.583	+.026
+ Similarity few-shot	0.610	+.027
+ Line-cited views	0.623	+.013
+ Hierarchy (Q1–Q4)	**0.706**	**+.083**

The four-way split does the heavy lifting: an 8-point jump, larger than every prompting trick combined. Few-shot barely helps until the structure is there.

AXIS 2 — Q3 PROMPT RUBRIC (Q3-STAGE MACRO-F1)

Q3 PROMPT	F1	Δ
Plain label list	0.603	–
+ Exclusion clauses	0.665	+.062
+ Contrastive examples	0.722	+.057
+ Chain-of-thought	**0.771**	**+.049**

Telling the judge when *not* to apply a label, plus near-miss examples, nearly doubles the score on the four reasoning-error labels. Each change compounds.

EXAMPLE

How our pipeline labelled the *Annual Report* failure.

Trace `multi_turn_base_6` · BFCL · GPT-5. The judge walks Q1 → Q2 and submits a verdict citing specific lines.

TRACE

```
L49 write Earning, Expenditure, Name
L54 agent:
echo("Name\n Earning\n Expend.")
L156 valid: False
L167 truth: single line, orig. order
```

→

Q1 Did the agent finish?

Reads L156 + L54. Agent *did* call `echo` and produce a final state. The state just did not match.

→ Unsuccessful Completion

→

Q2 Why was it wrong?

Compares L54 vs L167. Fields are right; order and line breaks are not. Not a chat-format issue, so Formatting Error is ruled out.

→ Output State Mismatch

→

VERDICT

USC, OSM

`cites L49 · L54 L156 · L167`

Human match: ✓

DISCUSSION

Main Takeaways

1 **Structure matters more than prompting.** Just splitting the verdict into four staged questions raised micro-F1 from 0.56 to 0.71, before we tuned any prompt. The few-shot tricks added afterward were worth only 0.03 more.

2 **Once the structure is fixed, the rubric is the lever.** Telling the judge when *not* to apply a label, plus near-miss examples, took Q3 from 0.60 to 0.77 — a bigger jump than any architecture change.

3 **Some labels just won't budge.** We threw five different fixes at Skipped Sub-Task; every one landed at F1 ≈ 0.43. The judge keeps over-applying it, and prompt wording can't stop it.

PER-CLASS RESULTS

Where the judge wins and struggles.

F1 on the BFCL hold-out (35 traces, 142 labels).

TOP PERFORMERS

Class	F1	n
Agent Errors (parent)	1.00	35
Incomplete Execution	0.88	19
Unsuccessful Completion	0.83	16
Function Invoc. Error	0.80	3

WHERE IT STRUGGLES

Class	F1	n
Output State Mismatch	0.42	8
Skipped Sub-Task (plateau)	0.38	5
Function Docs Ignored (n=1)	0.00	1

NEXT STEPS

From a better judge to a scaled AFT study.

- **Generalise beyond BFCL** — extend trace adapters to AppWorld and MCP-Universe.
- **Cross-model evaluation** — re-run on Sonnet 4.5, Kimi K2, Qwen3-32B.
- **Detect Fabrication structurally.** Prompt-only fixes have plateaued; a check on tool-call arguments is next.

CITATIONS

[1] Cemri et al. (2025). *Why Do Multi-Agent LLM Systems Fail?* arXiv:2503.13657.

[2] Zhu et al. (2025). *Where LLM Agents Fail and How They Can Learn From Failures.* arXiv:2509.25370.

[3] Zhang & Agrawala (2026). *View-oriented Conversation Compiler for Agent Trace Analysis.* arXiv:2603.29678.